Rainbow Demons

An original, independently designed Eurorack instrument

Release	Platform	Format	Firmware status
v2.0.0 / 2026-08-13	Daisy Patch SM	Tape-delay PCB / mono	Built from verified baseline

What this package contains

- Final source, Makefile, startup file and successfully compiled flashable binary.
- Tape-delay main PCB and faceplate Gerbers plus editable PCB source.
- Required PCB modifications, BOM, build sequence, acceptance test and quick-start control map.

Critical proven modifications

Add 100k from the audio-input switched contact to GND. Move the repurposed Reset-trigger tip from B6 to B9/GATE_IN_2. Record is D1, Reset is D2, Record LED is B8 and Status LED is B7. Feedback uses CV_7/C8. Do not restore obsolete ADC_12, D7/D10-button or B5/B6-LED assignments.

Release behavior

- Mix is counter-clockwise dry and clockwise wet with exact digital endpoints.
- Tape provides lo-fi pitch/reverse delay and high-feedback pitch spirals.
- Slice captures up to 4 seconds; Scatter captures up to 8 seconds.
- Scatter heads span 1/8x to 8x (-3 to +3 octaves) and stop near noon.
- REC CV clock edges alternate start / stop-and-play / start-new capture.
- Slice Feedback controls auto-record density; clockwise makes windows more frequent and shorter.
- Changing Mode clears and invalidates the previous capture.

Independent-design disclaimer

Rainbow Demons is inspired by broad buffer-manipulation ideas associated with MTL ASM's Count to 5. It is not a clone, reproduction, port, or claim of exact pedal behavior. No original source code, firmware, schematics, PCB files, or proprietary design material were used. This project is not affiliated with or endorsed by MTL ASM.

1. Required tape-delay PCB changes

Change	Final connection	Reason
Input pulldown	100k: input switched/normal contact to GND	Stops floating open input
Reset trigger	Isolate original B6 trace; jack tip to B9	B6 is an output; B9 is GATE_IN_2
Record trigger	Jack tip to B10	Clock edges toggle start / stop capture
Buttons	Record D1; Reset D2; other side GND	Matches final panel/firmware
Mode toggle	Outer lugs D7/D10; common GND	Tape / Slice / Scatter
Direction	Outer lugs D3/D4; common GND	Reverse / variable / forward
Quantize	Outer lugs D5/D6; common GND	Semitone / free / octave
LEDs	B8 Record; B7 Status; each through 1k	Plain GPIO indicators

Final signal map

Function	Pin	Function	Pin
Time pot	A2 / ADC_9	CV1 Time	C5 / CV_1
Feedback pot	C8 / CV_7	CV2 Filter	C4 / CV_2
Mix pot	C9 / CV_8	CV3 Flutter	C3 / CV_3
Filter pot	A3 / ADC_10	CV4 Feedback	C2 / CV_4
Flutter pot	D9 / ADC_11	CV5	C6 / disabled
Mono input	B4	Main output	B2
Optional output	B1	Ground	Jack sleeves / commons

Leave disconnected

- B6 from the repurposed Reset trigger jack; B5 FREEZE LED; audio-input-right in the mono build.
- CV5-to-Mix modulation. The final firmware disables it to preserve exact Mix endpoints.

Unpowered checks

- No shorts among +12V, -12V, 5V, 3V3 and GND; red stripe reaches marked -12V.
- Input normal contact measures about 100k to GND; Reset tip reaches B9 and not B6.
- Toggle commons and jack sleeves reach GND; LEDs have correct polarity and series resistors.

2. Bill of materials

Qty	Part	Specification / note
1	Daisy Patch SM	DSP and conditioned Eurorack I/O
1	Tape-delay main PCB	Main Gerbers included
4	Patch SM sockets	1x10, 2.54 mm female headers
1	Power header + cable	2x5 keyed; 10-to-16-pin ribbon; red stripe -12V
5	Potentiometers + knobs	B10K linear; Time, Feedback, Mix, Filter, Flutter
3	Toggles	SPDT ON-OFF-ON, panel mount
2	Buttons	SPST-NO momentary, Record and Reset
2	LEDs + 1k resistors	Diffused 3 or 5 mm; Record and Status
1	100k resistor	0.25 W input switched-contact pulldown
9	Switched 3.5 mm jacks	5 CV, 2 trigger, mono input, mono output
1 opt.	Second output jack	Tip to B1 for mirrored output
1	Perfboard	2.54 mm isolated pad-per-hole if controls are hand-wired
1 lot	Wire / heat-shrink	26-28 AWG stranded; 22-24 AWG tinned bus wire
1 set	Insulated hardware	M2.5/M3 standoffs, washers, rack screws
1	Front panel	Matching faceplate Gerbers included

Optional hand-wiring filter

For long pot leads, add 1k in series with each of five wipers and optionally 10 nF from the Patch SM side of each resistor to GND. These parts reduce pickup; they are not required for basic operation.

Mechanical rules

- Dry-fit every control, carrier and perfboard before soldering; preserve rail and USB/BOOT clearance.
- Support carrier and perfboard with insulated hardware. Never let perfboard copper rest on the carrier.
- Verify exact pot shaft/bushing, toggle action and jack footprint against purchased parts.

3. Assembly, flashing and checkout

Step	Task	Action
1	Mechanical dry fit	Fit panel controls, carrier and brackets. Confirm rail and USB/BOOT access.
2	Power	Install keyed header with red stripe at -12V. Do not feed external 5V/3V3 into Patch SM outputs.
3	Analog wiring	Wire pots/CVs per page 2. Ground sleeves. Add the 100k input pulldown.
4	Audio/triggers	Input B4; output B2; Record trigger B10; isolate B6 and bodge Reset trigger to B9.
5	Digital wiring	Record D1, Reset D2, toggles D3/D4/D5/D6/D7/D10, LEDs B8/B7.
6	Inspect	Meter rails, grounds, pulldown, B9 bodge, LED polarity and adjacent-header shorts.

Flash

Enter DFU: hold BOOT, tap RESET, release BOOT. In Git Bash inside the firmware folder:

Use	Command
Flash supplied build	make program-dfu
Clean rebuild + flash	make clean && make && make program-dfu

Acceptance test

- Power from a test/current-limited supply; stop if anything heats or current is abnormal.
- Mix fully counter-clockwise: clean dry audio. Fully clockwise: wet only, no direct leakage.
- Slice: hold Record, capture a phrase, release and hear playback.
- Scatter: capture, raise Flutter for three heads; Time/Feedback/Filter reverse left, stop near noon, forward right.
- REC CV: pulse 1 starts, pulse 2 stops/plays, pulse 3 starts a fresh capture.
- Slice Feedback: CCW disables auto-record; clockwise makes random windows more frequent and shorter. Record or REC CV takes priority.
- Change Mode and return. The old phrase must not resume.

Binary identity

RainbowDemons.bin - final binary identity is listed in BUILD_MANIFEST.txt.
Size: 88,252 bytes
B5329B1F9805EFE256EF1C30324392757952D0C81F3856A0632B72B872BF06DA

4. Quick start

Global control	Action
Mode	Up Tape / center Slice / down Scatter
Direction	Outer positions force direction; center lets DIR knobs cross reverse-stop-forward
Quantize	Outer positions select semitone/octave; center is free
Mix	Counter-clockwise dry / clockwise wet
Record	Hold to capture in Slice/Scatter; release to play
REC CV	Rising edges alternate start / stop-and-play / start fresh
Reset	Tap restarts heads; hold 1.2 seconds erases

If an outer toggle label is reversed, exchange only that toggle's two outside wires.

Mode control map

Knob	Tape	Slice	Scatter
Time	Read-head speed/direction	Slice speed/direction	Head 1 speed/direction
Feedback	Feedback amount	Auto-record density	Head 2 speed/direction
Mix	Dry/delay	Dry/slices	Dry/multi-head loop
Filter	Lo-fi tone/bandwidth	Random start probability	Head 3 speed/direction
Flutter	Stepped 31 ms-8 s length	Grain/full-phrase length	1, 2 or 3 heads

Mode notes

- REC CV accepts a positive Eurorack clock (0V low, +5V high recommended); pulse width is ignored.
- If a capture reaches its 4/8-second limit, it stops automatically and the next pulse starts fresh.
- Tape: high Feedback plus Time away from unity creates cascading, unstable harmonized pitch spirals.
- Slice: capture is up to 4 seconds; Filter raises random slice jumps; Flutter changes slice length.
- Slice density windows: sparse 1.25-3.75 s, noon 0.3-0.9 s, dense 0.06-0.2 s. Record or REC CV disables automation until Reset/mode change.
- The REC input senses voltage, not cable insertion; a patched 0 V cable takes priority when its first rising edge arrives.
- Scatter: capture is up to 8 seconds; each head spans 1/8x-8x (-3 to +3 octaves), about 1x near 1:30.
- Changing Mode always clears captured audio.

Repeatable test setup

- Direction/Quantize centered; Time and Filter noon; Flutter 2 o'clock. Use Feedback noon in Tape/Scatter, fully CCW for manual Slice tests.
- Leave CV inputs unpatched and change only Mix while verifying dry/wet behavior.

5. Release files and fabrication notes

Folder	Contents	Use
Top level	Build guide, modifications, BOM, quick start, PDF	Workshop reference
firmware/	Source, Makefile, startup, binary, manifest	Rebuild or flash final firmware
pcb/	Main/faceplate Gerbers, source, base map	Fabrication files; apply page 2 bodges

Gerber warning

The included tape-delay PCB Gerbers reproduce the supplied V2 carrier layout. They do not include the 100k input pulldown, B9 Reset-trigger bodge or final Rainbow Demons toggle wiring as copper revisions. Apply those changes manually, or update the KiCad source and rerun DRC before ordering a native revision.

Firmware build environment

- Target Daisy Patch SM / STM32H750; internal flash address 0x08000000.
- The included Makefile checks libDaisy, CMSIS and DaisySP before compiling.
- On another computer, override LIBDAISY_DIR and DAISYSP_DIR rather than hard-editing source paths.

Final preservation checklist

- Keep the release ZIP unchanged as the versioned 2026-08-13 baseline.
- Make experimental firmware in a new folder; do not overwrite this binary/source pair.
- Photograph the B9 bodge, 100k pulldown, red-stripe orientation and Patch SM orientation for the build record.
- If controls are replaced, confirm pot polarity and exchange toggle outer wires as needed.

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